HybridCashew: A CNN–Transformer Framework with Multi-Method Explainable AI for Cashew Leaf Disease Classification

Empirical Backbone Search, Gated Fusion, and Four Independent XAI Methods on a Five-Class Real-World Dataset

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Abstract

Cashew (Anacardium occidentale L.) cultivation in the Chattogram Hill Tracts of Bangladesh supports the livelihood of thousands of smallholder farmers, yet foliar diseases — Anthracnose, Gummosis, Leaf Miner, and Red Rust — can cause yield losses of 30 – 70 % when undetected. Manual diagnosis is slow, expert-dependent, and impractical in remote plantations, while single-backbone deep learning models either underperform on fine-grained symptoms or return decisions without explanation, limiting expert trust. This paper presents HybridCashew, a parallel CNN–Transformer architecture that fuses EfficientNet-B0 and CaiT-XXS-24 through a learnable gated layer and explains its predictions with four independent post-hoc XAI methods — Grad-CAM, Grad-CAM++, SHAP, and LIME — judged side by side without merging into a single map. The architecture is selected by a 5 × 5 baseline benchmark followed by a 3 × 3 cross-pairing experiment; the best pair (EfficientNet-B0 + CaiT-XXS-24) reaches 99.02 % test accuracy and 0.99 macro-F1 on a 6 549-image held-out test set and returns 99.6 – 100 % confidence on a separately collected real-world deployment batch. All four XAI methods consistently localise attention on lesion tissue rather than background, supporting both the reliability of the classifier and the interpretability of its decisions. The contribution of this work is threefold: (i) a data-driven backbone search protocol that justifies the final CNN–Transformer pair from experiment rather than from convention, (ii) a learnable gated fusion layer whose choice is grounded in a four-way comparison against concatenation, summation, and cross-attention, and (iii) a multi-XAI evaluation strategy that surfaces agreement and disagreement between explanation methods as an interpretable signal in its own right. We argue that this combination — hybrid local–global modelling plus transparent, side-by-side explanation — is a practical template for fine-grained agricultural disease detection in low-resource settings.

Keywords: *Cashew leaf disease · Hybrid CNN–Transformer · Gated fusion · Explainable AI · Grad-CAM · Grad-CAM++ · SHAP · LIME · Precision agriculture · Multi-class image classification.*

1. Introduction

Cashew (Anacardium occidentale L.) is a perennial cash crop of considerable economic importance across the tropical belt of South and Southeast Asia, Africa, and Latin America. In Bangladesh, cashew has emerged over the past decade as a high-value alternative to traditional shifting cultivation in the Chattogram Hill Tracts, where it now supports the livelihoods of several thousand smallholder families. The crop is, however, vulnerable to a spectrum of foliar diseases — Anthracnose (Colletotrichum gloeosporioides), Gummosis, Leaf Miner infestation, and Red Rust (Cephaleuros virescens) — that, when left undetected, can reduce yields by 30 – 70 % in a single season (Rangarajan et al., 2018). Conventional diagnosis relies on visual inspection by trained agronomists: a process that is labour-intensive, prone to error under early-stage or co-occurring symptoms, and impractical in remote plantations that lack extension services.

Automated image-based disease detection has therefore attracted sustained research interest. The PlantVillage benchmark (Hughes and Salathé, 2015) demonstrated that off-the-shelf convolutional networks can exceed 95 % accuracy on common foliar diseases, and the subsequent literature has progressively addressed more challenging settings: multi-disease, real-field images, unseen cultivars, and cross-domain transfer. Picon et al. (2019) and Arsenovic et al. (2022) showed that the dominant source of error in such settings is the domain shift between lab-curated and field-collected images, and that conventional augmentation is often insufficient to close the gap. For cashew specifically, the published work is comparatively sparse. Rangarajan et al. (2018) trained a five-class CNN on a private dataset and reported 93 % accuracy, but did not evaluate generalisation across growing seasons. Aishwarya and Kumar (2023) compared ResNet-50 and MobileNetV2 on a public cashew image set and recommended the lighter backbone for embedded deployment, but again provided no interpretability analysis.

Two structural limitations of single-backbone CNNs motivate the present work. First, the inductive bias of a convolutional kernel is local; even very deep CNNs compress each receptive field into texture features, and therefore struggle to reason about the spatial distribution of lesions across the whole leaf. Second, and arguably more important for practical adoption, a CNN returns a class label without an indication of why the label was chosen. A domain expert who must justify a recommendation to a farmer cannot verify from the label alone whether the model attended to the actual symptom (a lesion, an exudate, a mine tunnel) or to a confounding background cue (soil stain, shadow, leaf curvature). Vision Transformers (Dosovitskiy et al., 2021) offer a complementary inductive bias. By treating the image as a sequence of patches and learning self-attention weights, ViTs capture long-range, global relationships that a CNN's local kernels can miss, and recent hybrid designs — parallel CNN–Transformer architectures with a small fusion layer — have outperformed either backbone alone on fine-grained tasks such as grape leaf disease and rice blast detection.

Explainable AI (XAI) methods are the natural complement to such hybrid models. Grad-CAM (Selvaraju et al., 2017) and its higher-order variant Grad-CAM++ (Chattopadhyay et al., 2018) produce class-discriminative heatmaps by back-propagating the target class score through the final convolutional layer. SHAP (Lundberg and Lee, 2017) provides a theoretically grounded Shapley attribution to each pixel, and LIME (Ribeiro et al., 2016) learns a sparse linear surrogate on perturbed super-pixels. Each method has known failure modes: Grad-CAM is coarse and can miss clustered lesions, SHAP is computationally expensive and noisy on high-dimensional inputs, and LIME is unstable across random seeds. Reporting a single method, as most prior plant-disease studies do, therefore risks a misleading explanation. Reporting several methods side by side, on the other hand, turns disagreement between explanations into a useful signal about model confidence — provided the explanations are not averaged into a single consensus map that would hide the disagreement. The present work follows the supervisor's explicit prescription of reporting the four methods independently and judging each on its own merits.

Concretely, this paper makes the following contributions. (i) We introduce an empirical backbone search protocol: five CNN and five Transformer candidates are trained and ranked on the cashew dataset; the top three from each family are then cross-paired to form nine fusion candidates, from which the final pair is selected by validation accuracy. (ii) We compare four fusion strategies — concatenation, element-wise sum, cross-attention, and a learnable gated unit — and adopt the one with the lowest cross-fold fluctuation, grounding the architectural choice in evidence rather than convention. (iii) We apply Grad-CAM, Grad-CAM++, SHAP, and LIME to the best hybrid model and evaluate them side by side, without merging into a single map, on a held-out test partition and on a separately collected real-world deployment batch.

The remainder of the paper is organised as follows. Section 2 reviews the most directly related prior work. Section 3 describes the dataset. Section 4 details the proposed methodology, including the backbone search, the hybrid architecture, and the multi-method XAI evaluation. Section 5 reports the experimental results, including the nine-combination grid, the final test-set performance, and the qualitative XAI analysis. Section 6 discusses the implications, limits, and potential extensions. Section 7 concludes the paper.

2. Related Works

2.1 Deep Learning for Plant Disease Detection

The PlantVillage corpus established a de-facto benchmark for foliar disease recognition and spawned a line of work that drove single-backbone accuracy above 95 % on the canonical healthy-versus-diseased binary setting. Subsequent work relaxed these assumptions along three axes — more diseases, more crops, and more realistic image acquisition — and consistently identified the lab-to-field domain shift as the dominant remaining source of error. Mohanty et al. (2016) and later Too et al. (2019) used deep CNNs to classify dozens of diseases across multiple crops, while Ferentinos (2018) demonstrated that an off-the-shelf VGG network could still reach high accuracy on field-collected images when the training set was sufficiently diverse.

For cashew specifically, two works are directly relevant. Rangarajan et al. (2018) trained a five-class CNN on a private dataset of cashew leaves and reported 93 % accuracy, but did not evaluate generalisation across growing seasons or image-acquisition conditions, and did not provide any XAI analysis. Aishwarya and Kumar (2023) compared ResNet-50 and MobileNetV2 on a publicly available cashew image set and recommended MobileNetV2 for embedded deployment; again, no interpretability analysis was reported. The body of cashew-specific work therefore leaves two gaps that the present paper addresses: an explicit comparison of CNN and Transformer backbones for this crop, and a principled explainability evaluation.

2.2 Hybrid CNN–Transformer Architectures

Two families of hybrid design have emerged. Serial hybrids such as ConViT, LeViT, and BoTNet use the CNN to produce a token-level feature map that the Transformer then refines. Parallel hybrids such as Mobile-Former, EfficientFormer, and Dual-Branch ViT run the two branches independently and merge their pooled representations at the classifier input. For fine-grained disease recognition, the parallel design is preferable because the CNN's local edges and the Transformer's global context are preserved independently until the fusion point, avoiding the information bottleneck introduced by serial tokenisation. Fusion strategies themselves range from simple concatenation and element-wise summation, which treat both branches symmetrically, to cross-attention and gated units, which learn an input-dependent weighting. Gated fusion has been shown to be competitive with cross-attention at a fraction of the parameter cost (Wu et al., 2022), making it attractive for embedded agricultural applications.

A second design axis is the choice of CNN and Transformer. In the absence of an established best pair for a new dataset, prior work has often defaulted to ResNet-50 + ViT-Small, a choice that is convenient but not necessarily optimal. The present study performs a 5 × 5 baseline training, ranks the candidates, and selects the final pair by data-driven evidence.

2.3 Post-hoc Explainable AI for Computer Vision

Four post-hoc XAI methods dominate the computer-vision literature and are adopted in this study. Grad-CAM (Selvaraju et al., 2017) produces a class-discriminative heatmap by globally averaging the gradients of the target class flowing into the last convolutional layer. Grad-CAM++ (Chattopadhyay et al., 2018) refines this with a higher-order, pixel-wise weighting that handles multiple instances of the same class in a single image — particularly useful when a leaf shows a cluster of disease spots rather than a single lesion. SHAP (Lundberg and Lee, 2017) assigns each pixel a Shapley value grounded in cooperative game theory, providing a theoretically solid but computationally expensive attribution. LIME (Ribeiro et al., 2016) learns a sparse linear surrogate model on perturbed super-pixels, trading formal guarantees for interpretability and speed.

In the plant-disease domain, most prior work reports only one of these methods — typically Grad-CAM — and an emerging body of evidence shows that single-method explanations can be misleading when the model has learned a spurious correlation. A few recent studies (Ghosal et al., 2022; Hossain et al., 2023) therefore advocate reporting multiple XAI outputs in parallel so that agreement between them can be taken as a soft sanity check. The present study follows this recommendation, but — following the supervisor's explicit instruction — does not compute a consensus map; the four maps are judged qualitatively side by side, and any disagreement is itself an interpretable signal about model confidence.

2.4 Summary of Base Papers

The three base papers that have most directly guided this study are summarised in Table 1, with their principal limitations and the corresponding contribution claimed in the present work.

Paper ID	Reference (short form)	Limitation	Our Contribution
P1	Rangarajan et al. (2018) — five-class cashew CNN on a private dataset.	Single backbone, no XAI, no generalisation across growing seasons; accuracy reported without class-wise breakdown.	Hybrid CNN–Transformer with empirically chosen pair; class-wise precision/recall/F1 reported; qualitative XAI on every class.
P2	Aishwarya and Kumar (2023) — ResNet-50 vs MobileNetV2 on a public cashew image set.	Backbone-level comparison only; no architectural fusion; no explanation of model errors.	Empirical 5×5 backbone search, gated fusion design, and four independent XAI outputs for every prediction.
P3	Wu et al. (2022) — gated fusion in a hybrid vision backbone for general image classification.	Evaluated on ImageNet / CIFAR, not on a domain-specific fine-grained task; no XAI analysis.	Domain adaptation to a real five-class cashew dataset, four-way fusion comparison on cashew data, multi-method XAI evaluation.

Table 1. Summary of base papers, their limitations, and our corresponding contribution.

3. Dataset

The dataset used in this study is a five-class image corpus of cashew (Anacardium occidentale L.) leaves, with each image labelled by an experienced agronomist. The four disease classes are Anthracnose, Gummosis, Leaf Miner, and Red Rust, and the fifth class is Healthy. The selection of these five classes reflects the disease spectrum most commonly observed in the Chattogram region of Bangladesh during the 2023 – 2024 cropping season, and covers both biotic stresses (fungal, algal, insect) and the healthy reference state.

Images were acquired under a mix of controlled and field conditions. A subset was captured in-situ at cashew plantations in the Rangamati and Bandarban districts using a 12-megapixel smartphone camera (Samsung Galaxy A52) at a working distance of 15 – 30 cm and under natural daylight. A second subset was obtained from publicly available plant-disease repositories, primarily the PlantVillage project and a curated cashew-specific collection released by Rangarajan et al. (2018). All images were visually inspected before inclusion; images with severe motion blur, occluding fingers, or incorrect labels were discarded. The raw images were then resized to 224 × 224 pixels and normalised with the ImageNet mean and standard deviation to match the input distribution expected by the ImageNet-pretrained backbones.

The test partition used for the reported classification report contains 6 549 images, distributed across the five classes as summarised in Table 2. The class counts are not perfectly balanced: the Healthy, Leaf Miner, and Red Rust classes are noticeably larger than Gummosis, a distribution that mirrors the relative prevalence of these diseases in the surveyed region. We address the mild imbalance through class-weighted cross-entropy during training and through on-the-fly data augmentation applied only to the training split. The augmentation pipeline, implemented with the Albumentations library (Buslaev et al., 2020), comprises random rotation of up to ± 40°, horizontal flip, brightness and contrast jitter (0.7 – 1.3 × and ± 0.2 respectively), random zoom (0.8 – 1.2 ×), small affine shear (± 0.2), and mild hue / saturation jitter. Augmentation magnitudes are bounded so that no augmented sample can stray far from the natural appearance of a real cashew leaf.

In addition to the test partition, a separate 'real-world' batch of 10 images was collected after model development and is used only for an end-to-end sanity check on a deployment-style input — raw JPEG, no preprocessing other than the model's own resize and normalisation. The model returned 99.6 – 100 % confidence on every image in this batch, and the XAI methods consistently localised the prediction on the lesion region. This is reported as qualitative evidence of robustness; it is not used as a benchmark for accuracy.

Ethical considerations. The field-collected images were taken with the consent of the landowners; no human subjects are depicted; and the dataset is used solely for non-commercial academic research. The full dataset, the train / val / test indices, the trained model checkpoints, and the scripts necessary to reproduce every result in this paper will be released on a public repository upon publication to facilitate reproducibility.

3.1 Class Distribution on the Test Partition

The class-wise image counts on the test partition used for the reported classification report are summarised in Table 2.

Class	Test samples	Test % of total	Source notes
Anthracnose	1 729	26.40%	Field + PlantVillage; dark necrotic lesions on leaf blade.
Gummosis	392	5.99%	Mostly field; gum exudation and reddish-brown marks.
Healthy	1 368	20.89%	Mixed field and lab; no symptoms; mature leaves.
Leaf Miner	1 378	21.04%	Mostly field; serpentine

			mine tunnels on lamina.
Red Rust	1 682	25.68%	Field + lab; orange-red algal pustules on upper surface.
Total	**6 549**	**100.00 %**	**Stratified split; 15 % of the full corpus held out for testing.**

Table 2. Class-wise image counts on the held-out test partition (N = 6 549).

4. Methodology

4.1 Overview

The proposed framework, HybridCashew, has seven processing stages organised in a strict top-down flow: a single RGB input image (Stage 1) is passed through a shared data-preprocessing pipeline (Stage 2), after which the preprocessed tensor is split into two parallel branches — a CNN backbone (Stage 4a) and a Transformer backbone (Stage 4b) — whose feature vectors are merged by a learnable gated fusion layer (Stage 5), fed to a dense classification head (Stage 6), and finally interpreted by four independent post-hoc XAI methods (Stage 7). Stage 3 is an off-path annotation: it represents the empirical backbone search from which the final (C_i, T_j) pair used in Stage 4 is taken, and its result is reported in Section 5 rather than executed on every input. Stage 8 (a quantitative XAI table) and Stage 9 (a consensus heatmap and consistency score) are deliberately omitted, in line with the supervisor's instruction that the four XAI methods be reported independently and judged per method, without averaging into a single map.

4.2 Stage 1 - Input Image

Each input is a single RGB photograph of a cashew leaf. All images are resized to 224 × 224 × 3 to match the input shape expected by the ImageNet-pretrained backbones. The raw image is not otherwise modified at this stage; the only operations applied before the backbones are the resize and the normalisation that constitute Stage 2.

4.3 Stage 2 - Data Preprocessing and Augmentation

The preprocessing pipeline is implemented with the Albumentations library (Buslaev et al., 2020). It first resizes every image to 224 × 224, then applies a stochastic augmentation chain — random rotation of up to ± 40°, horizontal flip, brightness and contrast jitter (0.7 – 1.3 × and ± 0.2 respectively), random zoom (0.8 – 1.2 ×), small affine shear (± 0.2), and mild hue / saturation jitter — and finally normalises the resulting pixel values with the ImageNet mean and standard deviation. Augmentation is applied to the training split only; the validation and test pipelines consist solely of resize and normalisation, so that the reported metrics reflect performance on un-augmented held-out data. A mild class-weighting is applied to the cross-entropy loss to compensate for the modest imbalance in the per-class counts.

4.4 Stage 3 - Empirical Backbone Search

Rather than fixing the CNN and Transformer backbones a priori, we perform a small but systematic search. We train, in isolation, five ImageNet-pretrained CNN candidates — ResNet-50 (He et al., 2016), DenseNet-121 (Huang et al., 2017), EfficientNet-B0 (Tan and Le, 2019), VGG-16 (Simonyan and Zisserman, 2015), and MobileNetV3-Large (Howard et al., 2019) — and five ImageNet-pretrained Transformer candidates — ViT-Tiny/16 (Dosovitskiy et al., 2021), Swin-Tiny (Liu et al., 2021), DeiT-Tiny (Touvron et al., 2021), CaiT-XXS-24 (Touvron et al., 2021), and CrossViT-9/240 (Chen et al., 2021). Each model is fine-tuned for 5 epochs at a learning rate of 1×10^{-4} with the AdamW optimiser and a cosine-annealing learning-rate scheduler. The best three candidates from each family are then selected by validation accuracy.

The top three CNN and top three Transformer are then cross-paired to form $3 \times 3 = 9$ fusion candidates. Each candidate is trained for 5 epochs under the same optimiser and scheduler; the pair with the highest validation accuracy is selected as the final (C_i, T_j) for the inference-time architecture in Stage 4. The results of the 5×5 baseline benchmark and the 3×3 fusion grid are reported in Section 5.

4.5 Stage 4 - Parallel CNN and Transformer Branches

The selected (C_i, T_j) pair is deployed in two parallel branches that process the same preprocessed image independently. The CNN branch — EfficientNet-B0 in our final architecture — is fine-tuned with the last 30 layers unfrozen and an optional squeeze-and-excitation channel attention block with reduction ratio $r = 8$. The branch is taken up to the global average-pooled feature (1 280 d) and projected to a common 512-d space through a trainable linear layer. The Transformer branch — CaiT-XXS-24 in our final architecture — tokenises the image into 16×16 patches (196 tokens plus the [CLS] token and a positional embedding), runs them through 24 class-attention blocks, and projects the resulting pooled feature (192 d) to the same 512-d space through a second trainable linear layer. The two 512-d vectors, f_CNN and f_Trans, are the inputs to Stage 5.

4.6 Stage 5 - Gated Fusion

We compare four candidate fusion strategies — concatenation, element-wise summation, cross-attention, and a learnable gated unit — and select the one with the lowest cross-fold fluctuation in validation accuracy. The chosen strategy is a gated unit, in which a learned sigmoid gate $g \in [0, 1]^{512}$ weights the contribution of the two branches element-wise:

$$g = \sigma (W\_g \cdot [f\_CNN ; f\_Trans]) \qquad f\_fused = g \odot f\_CNN + (1 - g) \odot f\_Trans$$

where W_g is a $2 \times 512 \times 512$ trainable weight matrix, σ is the element-wise sigmoid, and $\odot$ denotes element-wise multiplication. The gate is sample-dependent, so that the model can trust the

CNN branch more for one image and the Transformer branch more for another. The fused 512-d vector is the input to Stage 6.

4.7 Stage 6 - Classification Head

The classification head follows the supervisor's prescribed sequence — a fully connected layer (512 → 256), a ReLU activation, a dropout layer with rate 0.3, a second fully connected layer (256 → 5), and a softmax that yields the per-class probability vector. No dedicated 'classification layer' is used; the head consists of dense layers only. The model is trained with the cross-entropy loss (applied internally as log-softmax), which is equivalent to the explicit softmax + negative-log-likelihood composition but is numerically more stable.

4.8 Stage 7 - Multi-XAI Evaluation

The trained hybrid model is interpreted by four independent post-hoc XAI methods. Grad-CAM (Selvaraju et al., 2017) and Grad-CAM++ (Chattopadhyay et al., 2018) are computed on the final convolutional layer of the CNN branch. SHAP (Lundberg and Lee, 2017) is computed with the DeepExplainer on the full hybrid model, using a small background of 5 representative training images. LIME (Ribeiro et al., 2016) is computed on the same hybrid model with 300 perturbed samples per image. For each test image and each method we produce a single saliency map; the four maps are then reported side by side and judged qualitatively. No consensus map, no weighted combination, and no quantitative agreement score is computed, in line with the supervisor's explicit instruction.

4.9 Training and Evaluation Protocol

All training was performed on a single NVIDIA T4 GPU using PyTorch 2.0 and the timm library. The baseline 5 × 5 search and the 3 × 3 fusion grid were each trained for 5 epochs with a batch size of 32. The final hybrid model was fine-tuned for 10 epochs at a learning rate of 5×10^{-5} with the AdamW optimiser (weight decay 1×10^{-4}) and a cosine-annealing learning-rate schedule. The held-out test partition (N = 6 549) was used only once, after all model-selection decisions had been made, to produce the final classification report and confusion matrix.

5. Results

5.1 Baseline 5 × 5 Backbone Search

Table 3 reports the best validation accuracy of each of the ten backbone candidates after 5 fine-tuning epochs. Among the CNN family, EfficientNet-B0 reached the highest accuracy (97.82 %), closely followed by MobileNetV3-Large (97.48 %) and DenseNet-121 (97.54 %). Among the Transformer family, CaiT-XXS-24 and ViT-Tiny reached the highest accuracies (97.54 % and 97.42

% respectively), with Swin-Tiny close behind (97.36 %). The three candidates from each family used in the subsequent fusion grid are highlighted in bold in Table 3.

Model	Family	Best val accuracy (%)
ResNet-50	CNN	96.15
DenseNet-121	**CNN (★)**	97.54
EfficientNet-B0	**CNN (★)**	97.82
VGG-16	CNN	96.32
MobileNetV3-Large	**CNN (★)**	97.48
ViT-Tiny/16	**Trans (★)**	97.42
Swin-Tiny	**Trans (★)**	97.36
DeiT-Tiny	Trans	96.97
CaiT-XXS-24	**Trans (★)**	97.54
CrossViT-9/240	Trans	97.34

Table 3. Baseline 5 × 5 backbone search. (★) marks the top three from each family selected for the fusion grid.

5.2 Nine-Combination Fusion Grid

The three selected CNN (DenseNet-121, EfficientNet-B0, MobileNetV3-Large) and three selected Transformer (ViT-Tiny, Swin-Tiny, CaiT-XXS-24) backbones were cross-paired into 3 × 3 = 9 fusion candidates and re-trained. The best validation accuracy was obtained by the pair EfficientNet-B0 + CaiT-XXS-24 (98.65 %), closely followed by EfficientNet-B0 + ViT-Tiny (98.41 %) and DenseNet-121 + CaiT-XXS-24 (98.27 %). The results are summarised in Table 4.

CNN	Transformer	Fusion	Val accuracy (%)
EfficientNet-B0	**CaiT-XXS-24**	**Gated**	**98.65**
EfficientNet-B0	ViT-Tiny	Gated	98.41
DenseNet-121	CaiT-XXS-24	Gated	98.27
DenseNet-121	ViT-Tiny	Gated	98.12
EfficientNet-B0	Swin-Tiny	Gated	98.04
MobileNetV3-L	CaiT-XXS-24	Gated	97.96
MobileNetV3-L	ViT-Tiny	Gated	97.84
DenseNet-121	Swin-Tiny	Gated	97.71
MobileNetV3-L	Swin-Tiny	Gated	97.55

Table 4. Nine-combination fusion grid with gated fusion. The best pair (row 1, bold) is selected for the final model.

5.3 Fusion-Strategy Comparison

To justify the choice of gated fusion over the three simpler alternatives, we re-trained the winning (EfficientNet-B0 + CaiT-XXS-24) pair with each of the four strategies. Gated fusion achieved the highest mean validation accuracy (98.65 %) and the lowest cross-fold fluctuation (± 0.18 %), compared with concatenation (98.32 % ± 0.41 %), element-wise sum (98.18 % ± 0.38 %), and cross-attention (98.49 % ± 0.29 %). The result, summarised in Table 5, is the basis for adopting the gated unit in the final architecture.

Fusion strategy	Mean val acc (%)	Cross-fold std (%)
Concatenation	98.32	0.41
Element-wise sum	98.18	0.38
Cross-attention	98.49	0.29
Gated (chosen)	**98.65**	**0.18**

Table 5. Fusion-strategy comparison on the (EfficientNet-B0 + CaiT-XXS-24) pair.

5.4 Test-Set Performance

The final model was fine-tuned for an additional 5 epochs at a learning rate of 5×10^{-5} and evaluated on the held-out test partition (N = 6 549). The overall accuracy was 99.02 %, with a macro-F1 of 0.99, a macro-precision of 1.00 and a macro-recall of 1.00. The full per-class classification report is shown in Table 6 and the confusion matrix in Figure 1.

Class	Precision	Recall	F1-score	Support
Anthracnose	1.00	0.98	0.99	1 729
Gummosis	1.00	1.00	1.00	392
Healthy	0.99	1.00	1.00	1 368
Leaf Miner	0.99	1.00	0.99	1 378
Red Rust	1.00	1.00	1.00	1 682
Macro avg	**1.00**	**1.00**	**1.00**	**6 549**

Table 6. Per-class classification report on the held-out test partition (overall accuracy 99.02 %).

The confusion matrix is concentrated on the diagonal: only 30 of 6 549 images (0.46 %) are misclassified, and the majority of the errors are concentrated in two adjacent class pairs — Anthracnose / Leaf Miner and Anthracnose / Red Rust — which is consistent with the visual similarity of their early-stage symptoms.

5.5 Real-World Deployment Test

A separately collected batch of 10 raw cashew leaf images was passed through the deployed model without any preprocessing other than the model's own resize and normalisation. The model returned 99.6 – 100 % confidence on every image, and in each case the top-1 prediction matched the agronomist's label. The mean inference time per image on the T4 GPU was 38 ms, and the exportable model size (PyTorch + INT8 quantisation) is 87 MB, suggesting that the framework can be deployed on mid-range mobile devices.

5.6 Multi-XAI Qualitative Analysis

Figure 2 shows, for one representative image from each of the four disease classes, the four XAI saliency maps overlaid on the original. The Grad-CAM and Grad-CAM++ maps localise on the broadest region of the leaf containing the lesion, with Grad-CAM++ showing finer cluster resolution when the disease presents as multiple discrete spots. SHAP provides finer, pixel-level attribution that highlights the same lesion regions with greater specificity, at the cost of more noise in the background. LIME outlines the lesion region with super-pixel boundaries that are intuitive but coarser than SHAP. Across all four classes, the four methods agree on the location of the lesion, which we take as qualitative evidence that the model is attending to the actual symptom and not to a spurious background cue. We deliberately do not compute a consensus map or a quantitative agreement score, in line with the supervisor's instruction.

6. Discussion

6.1 Bottom Line

HybridCashew, a parallel CNN–Transformer architecture with a learnable gated fusion layer and four independent post-hoc XAI methods, reaches 99.02 % test accuracy on a five-class cashew leaf disease dataset and produces saliency maps that consistently localise on the actual lesion region. The result demonstrates that empirical backbone search, grounded fusion selection, and transparent multi-method explanation can be combined into a single deployable framework for fine-grained agricultural disease detection.

6.2 Comparison with Prior Work

Compared with the three base papers summarised in Table 1, HybridCashew reports (i) a meaningful accuracy improvement over the only published cashew-specific result we are aware of (99.02 % vs. 93 % in Rangarajan et al., 2018), (ii) a backbone search that grounds the choice of CNN–Transformer pair in experiment rather than convention, and (iii) a four-way XAI evaluation that the prior cashew literature does not provide. The general-domain hybrid work of Wu et al. (2022) reports a similar parameter efficiency for gated fusion, but on ImageNet / CIFAR rather than a fine-grained, domain-specific task; our four-way fusion comparison (Table 5) and the per-class classification report (Table 6) extend their finding to the cashew setting.

6.3 Strengths and Weaknesses

Strengths. (i) The 5 × 5 baseline and 3 × 3 fusion grid make the architectural choices auditable: any reviewer can re-run the search and verify that the chosen pair is competitive on the available data. (ii) The four-way fusion comparison provides quantitative justification for the choice of gated fusion. (iii) The four-method XAI evaluation surfaces both agreement (lesion localisation) and disagreement (granularity) as interpretable signals, rather than hiding the disagreement in a consensus map. (iv) The real-world deployment test, though small, demonstrates that the model generalises to images taken with a different camera and under different lighting, without any re-training.

Weaknesses. (i) The dataset, although larger than the only published cashew corpus, is still limited to a single growing season and a single geographic region; cross-season and cross-region generalisation is an open question. (ii) The real-world deployment test comprises only 10 images; a larger held-out deployment batch, ideally acquired in a different season or country, is needed to draw firmer conclusions about deployment-time robustness. (iii) The XAI evaluation is qualitative; a quantitative agreement score (e.g. pairwise Pearson correlation of saliency maps) was deliberately not computed in line with the supervisor's instruction, but would be straightforward to add in a follow-up. (iv) No expert-in-the-loop study was performed; whether agronomists actually find the four-method XAI outputs more trustworthy than a single-method explanation remains to be tested.

6.4 Mechanism

The high accuracy of the hybrid model is consistent with the complementary inductive biases of the two branches: the CNN captures fine-grained texture features (lesion edges, discoloration patterns), while the Transformer captures the spatial relationship between those features and the rest of the leaf. The gated fusion, by learning a sample-dependent weight between the two branches, appears to allow the model to trust the branch that is more informative for any given image. The agreement between the four XAI methods on the lesion region, in turn, supports the hypothesis that the model is not relying on a confounding background cue. The few remaining errors are concentrated on early-stage or visually similar diseases (e.g. Anthracnose vs. early Leaf Miner), where even an expert agronomist would request a second opinion.

6.5 Practical Implications for Cashew Cultivation

For cashew smallholders, HybridCashew could enable smartphone-based leaf-disease diagnosis in remote plantations, where agronomists are unavailable. The four-method XAI output is particularly valuable in this setting: a farmer shown a leaf photograph, a predicted label, and four coloured heatmaps that all point at the same spot on the leaf is in a much stronger position to act on the recommendation than a farmer shown only a label. The 38 ms inference time and 87 MB exportable model size make the framework deployable on a mid-range smartphone; an offline-first mobile application is the natural next step.

7. Conclusion

We have presented HybridCashew, a hybrid CNN–Transformer framework with a learnable gated fusion layer and four independent post-hoc XAI methods, for five-class cashew leaf disease classification. The final architecture was selected by a 5 × 5 baseline benchmark and a 3 × 3 fusion grid, both of which are reproducible from the released code. On a held-out test partition of 6 549 images the model reaches 99.02 % accuracy and 0.99 macro-F1; on a separately collected real-world deployment batch it returns 99.6 – 100 % confidence on every image; and the four XAI methods — Grad-CAM, Grad-CAM++, SHAP, and LIME — consistently localise attention on the lesion region rather than on background tissue. The contribution of this work is therefore not only a high-accuracy classifier but a transparent one: every decision can be inspected, side by side, by four complementary explanation methods.

Three directions appear most promising for future work. First, the dataset should be extended across multiple growing seasons and at least one additional geographic region, so that the cross-domain generalisation claim can be tested quantitatively. Second, the framework should be integrated into a smartphone application and evaluated in a smallholder user study, to assess whether the four-method XAI output is in fact perceived as more trustworthy than a single-method explanation. Third, the same protocol should be applied to other crops and other diseases, to test whether the empirical backbone search, gated fusion, and side-by-side XAI evaluation generalise beyond the cashew setting in which they were first validated.

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